

On the Emergence and Initial Flight Behaviour of the Mountain Pine Beetle, *Dendroctonus ponderosae*, in Eastern Washington¹

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With 4 Figures

Abstract

Emergence from ponderosa pine and initial flight behaviour of the bark beetle, *Dendroctonus ponderosae* were studied under field conditions. Hourly emergence increased as ambient temperatures rose from 20 to 30° C but declined markedly above 30° C. Essentially no emergence activity occurred on days when temperatures remained below 16° C. Under optimal temperature conditions (25–30° C), peak hourly emergence did not synchronize with maximum daily temperatures, suggesting that emergence is controlled rhythmically. The average sex ratio of emergent adults was 1 male to 1.4 females, although diurnal variation was considerable. In flight tests, newly emergent adults generally flew with the wind in the absence of pheromones. In contrast, in the presence of synthetic trans-verbenol and fresh host material, most male and female beetles flew directly towards the attraction source against the wind. Flight exercise was not a prerequisite to olfactory response. Flight height above the ground appeared to be independent of either ambient temperature or wind speed. Although the proportion of beetles which responded to attractants decreased with increasing wind speeds, several beetles flew against winds exceeding 7.5 km/h.

Introduction

Host selection by the mountain pine beetle, *Dendroctonus ponderosae* Hopkins (Coleoptera: Scolytidae), is characterized by a sequence of stimulus-response mechanisms which insures allocation of new breeding material. In recent years, various aspects of the host selection behaviour have been investigated, both in the laboratory (McCAMBRIDGE 1971; SHEPERD 1966; WATSON 1970) and in the field (McCAMBRIDGE 1964, 1967; PITMAN 1971; PITMAN and VITÉ 1969; REID 1962). Nevertheless, certain aspects of the host selection sequence remain poorly understood. This paper describes field studies conducted during August, 1971, to clarify two initial phases: (1) diurnal emergence of adults and (2) initial flight as influenced by environmental factors and volatile attractants.

¹ Study supported by the McIntire-Stennis program and a Weyerhaeuser fellowship; administered by the Institute of Forest Products, University of Washington.

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Materials und Methods

The studies were conducted at a forested site known locally as "Elk Heights" (elevation 860 meters), situated 23 kilometers west of Ellensburg, Washington. The forest consisted primarily of ponderosa pine, *Pinus ponderosa* Lawson, with mixtures of Douglas-fir, *Pseudotsuga menziesii* (Mirb.) Franco, and grand fir, *Abies grandis* (Douglas) Lindl. Mountain pine beetles for field tests were obtained from a large bark beetle population active in the second-growth stands (50–60 years old) of ponderosa pine. The seasonal flight of *D. ponderosae* in the study area began in mid-July and terminated in early September. Accordingly, it should be recognized that beetles used in the study represented late-emerging populations.

To study beetle emergence, infested bolts of ponderosa pine were placed in a $2.5 \times 2.5 \times 2.0$ meter screened cage and maintained under field conditions of partial shade. Emergent beetles were sampled daily from August 9 to August 16, 1971, and hourly (6 a.m. – 8 p.m. Pacific Standard Time) from August 17–26. Beetles collected from the top and sides (above 1.5 m) of the cage were counted and sexed by the presence or absence of a stridulating mechanism, as described by MCCAMBRIDGE (1962). Hourly emergence activity was correlated with temperature, the latter monitored with a mercury thermometer situated within the emergence cage.

During the period of August 17 to 26, following each hourly collection, the initial flight behaviour of individual beetles was observed in a nearby flight arena. The arena consisted of a four-sided ($5.5 \times 5.7 \times 7.7 \times 8.9$ meter) cheesecloth enclosure attached at the corners to trees to a height of 4.3 meters. The floor was cleared of secondary vegetation and covered with a sheet of white muslin to facilitate observation and recovery of low flying beetles.

Flight by the newly emergent beetles was tested under two conditions: 1. for six days in the apparent absence of nearby natural or synthetic attractants, and 2. for three days in the presence of two olfactometers, each baited with a 0.6 meter long ponderosa pine bolt and synthetic attractants (*trans*-verbenol and an active host terpene [myrcene], dispensed from capillary polyethylene tubing and a 2-dram polyethylene vial, respectively) (see PITMAN 1971). The olfactometers, of the type described by CADE et al. (1970), were placed upwind near two corners of the arena 3.5 m from the arena center. Beetles were placed individually on a 0.7 meter square muslin platform staked 0.3 m above the ground in the center of the arena. To induce flight, each beetle was propelled into the air by snapping the underside of the platform. However, during the last three days of the study, freshly emerged beetles of both sexes were left undisturbed on the platform for 10 minutes and allowed to take to flight unassisted; beetles remaining after 10 minutes were snapped into the air.

Within the arena during the flight of each beetle, recordings were made of wind direction with two wind vanes and wind speed by means of a Hastings Air Meter, Model RB-1. The airborne beetles were watched throughout their initial flight, and the height at which they hit or passed over the arena was recorded.

In an additional test, a large number of beetles was collected off the arena's sides over the period August 24–26 and induced to fly in an open situation. To provide a source of attraction, a baited olfactometer similar to those used within the arena was placed upwind (3.5 m) from the release point. Individual beetles were placed in a petri-dish and flicked into the air to stimulate flight. For each beetle tested, recordings were made of the flight direction in relation to the olfactometer, as well as the wind direction, wind speed, and ambient temperature.

Results and Discussion

Emergence Activity: From August 17–26, total daily emergence of male and female *D. ponderosae* correlated directly with maximum daily temperatures on those days when temperatures did not exceed 30°C (Fig. 1). On August 19 and 26, when temperatures rose above 30°C , total numbers of emerging beetles appeared to diminish. Of particular interest was the sharp decline in emergence activity which occurred when a cold front was experienced, and

the immediate increase in numbers as the weather became warmer. As shown in Fig. 1, a brief cold spell apparently had little or no permanent effect on bark beetle emergence. It is known, however, that prolonged cold periods in

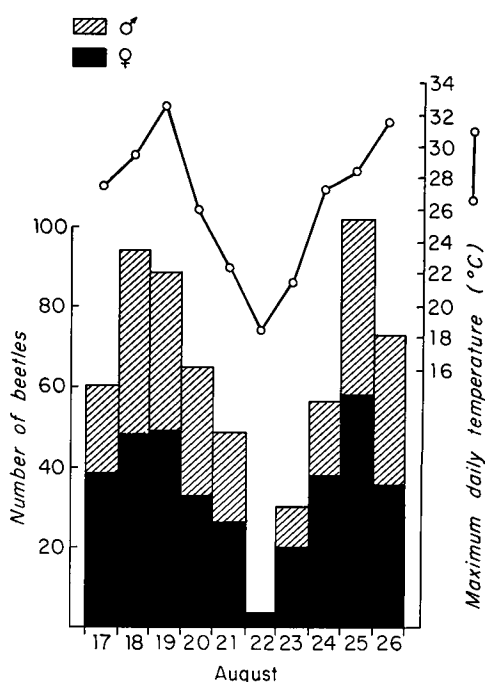


Fig. 1. Total diurnal emergence of *Dendroctonus ponderosae* in relation to maximum daily temperature

early summer may delay initiation of seasonal bark beetle emergence (McCAMBRIDGE 1964; REID 1962); whereas, later in the flight period, cold temperatures may completely curtail emergence for the season (MILLER and KEEN 1960). Indeed, FURNISS (1965) recorded a case in which populations of the Douglas-fir beetle, *D. pseudo-tsugae* Hopkins, in southern Utah, failed to emergence as predicted and passed two winters in the same trees. Abnormally low seasonal temperatures were believed to be responsible.

Although the sex ratio of emerging *D. ponderosae* was found to vary considerably from day to day, no consistent change was recorded as the season progressed (Table 1). Of 1031 adults collected from the emergence cage, 430 were males and 601 females, giving an overall male to female ratio of 1:1.4. This ratio is considerably

lower than that reported by REID (1958) of 1:2 or by MCGHEHEY (1969) of 1:1.8 for *D. ponderosae* reared from lodgepole pine (*P. contorta* Doug. var. *latifolia* Engelm.) in British Columbia. On the other hand, a 1:1 ratio has been observed for mature broods in western white pine (*P. monticola* Doug.) in Idaho (DE LEON et al. 1934). These differences suggest that perhaps the sex ratio varies significantly with geographical location or among different host species.

Patterns of average hourly emergence corresponding to four distinct regimes of maximum daily temperature are shown in Fig. 2. On days when maximum temperatures remained within optimal limits (25–30°C), emergence activity (Fig. 2A) began when temperatures rose above 20°C, then increased in the morning hours with increasing temperatures, and ceased in the afternoon at about the same threshold temperature. Peak emergence occurred between 11 a. m. and 2 p. m. when 61 percent (232 out of 379) of the beetles were collected, even though maximum temperatures were recorded somewhat later (2–4 p. m.). REID (1962) reported a similar pattern of *D. ponderosae* emergence for a single day in the East Kootenay region of British Columbia. The lack of synchronization between peak emergence and maximum temperatures (Fig. 2A) suggests that emergence activity is rhyth-

Table 1

Date (1971)	Number of Males	Number of Females	Total	Ratio Male : Female
August 7	45	62	107	1 : 1.4
9	12	35	47	1 : 2.9
10	25	49	74	1 : 2.0
11	15	41	56	1 : 2.8
12	30	37	67	1 : 1.3
16	8	15	23	1 : 1.9
17	23	38	61	1 : 1.6
18	46	48	94	1 : 1.0
19	40	49	89	1 : 1.2
20	32	33	65	1 : 1.0
21	23	26	49	1 : 1.1
23	10	20	30	1 : 2.0
24	19	38	57	1 : 2.0
25	44	58	102	1 : 1.3
26	38	35	73	1 : 0.9
30	20	17	37	1 : 0.8
Total	430	601	1031	1 : 1.4

mically controlled. WATSON (1970) recently reported similar findings for *D. ponderosae* reared from lodgepole pine under conditions of constant temperature and darkness.

The emergence pattern for August 7 (Fig. 2B) again illustrates the sharp decline in activity which occurred when ambient temperatures exceeded 30°C.

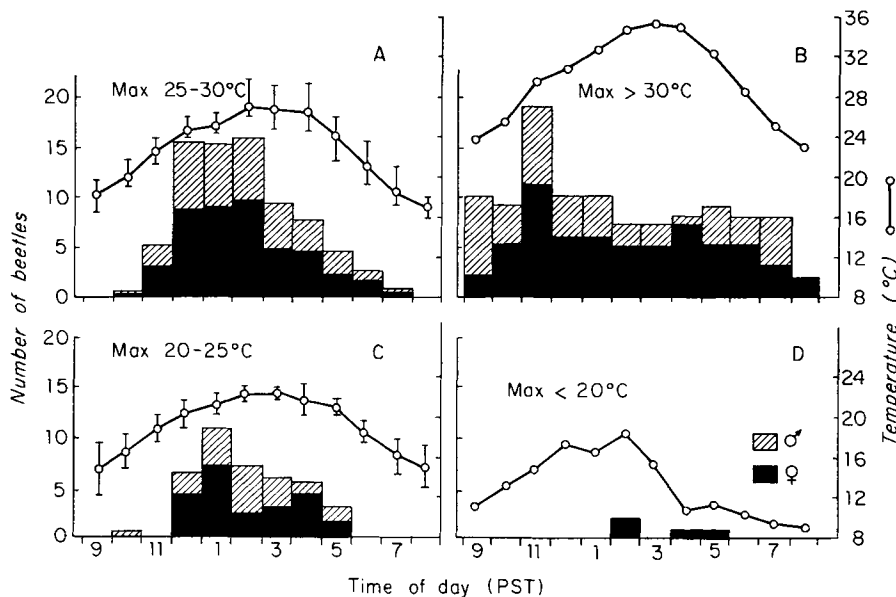


Fig. 2. Hourly emergence of *Dendroctonus ponderosae* from ponderosa pine in relation to 4 regimes of maximum daily temperatures. A. Average of 5 days – B. August 7, 1971 – C. Average of 2 days – D. August 22, 1971. Vertical lines represent temperature ranges

In contrast to days of lower temperatures, only 21 percent of the daily total emerged between 11 a. m. and 2 p. m. In addition, the duration of diurnal emergence was prolonged over a 12-hour period. A bimodal emergence pattern, when temperatures exceed 22° C, has been reported for female *Ips confusus* Lec. reared from ponderosa pine (CAMERON and BORDEN 1967). However, no definite bimodal pattern at high temperatures nor variation with sex was evident in our studies.

On colder days, when temperatures remained below 25° C (Fig. 2C, D), emergence was limited to a few hours during the day when temperatures were optimal. If temperatures fail to exceed approximately 16° C, emergence would be essentially curtailed for that day.

Initial Flight Behaviour in the Arena: During the eight days when temperatures were favourable, a high percentage of the newly emerged beetles flew (520 out of 592 or 87 %). Of these, 106 out of 220 or 40 percent of them flew of their own accord when left on the platform for up to 10 minutes. Essentially, all the beetles flew in the presence of wind and only 21 of 522, or approximately 4 percent, were lost or unobserved while in flight.

The numbers of beetles flying either with, against, or across the prevailing wind are given in Table 2. In the absence of baited olfactometers, most

Table 2

Initial Flight response of newly emergent *Dendroctonus ponderosae* beetles to wind when tested under two conditions within an arena

Date (1971)	Sex	With Wind	Against Wind	Across Wind	Up	Total
A. In absence of olfactometers baited with attractants						
August 17	♀	22	4	5	6	37
	♂	8	3	2	3	16
18	♀	16	14	7	1	38
	♂	19	13	4	2	38
19	♀	23	14	6	1	44
	♂	26	2	3	2	33
20	♀	10	14	1	0	25
	♂	14	14	2	0	30
21	♀	16	7	0	0	23
	♂	13	2	0	1	16
Total	♀	87	53	19	8	167
	♂	80	34	11	8	133
Total Grand	♀ + ♂	167	87	30	16	300
B. In presence of olfactometers baited with attractants						
August 24	♀	21	8	5	0	34
	♂	7	7	1	0	15
25	♀	3	35	8	2	48
	♂	9	23	6	0	38
26	♀	8	20	5	0	33
	♂	12	15	4	0	31
Total	♀	32	63	18	2	115
	♂	28	45	11	0	84
Total Grand	♀ + ♂	60	108	29	2	199

beetles (56 %) flew with the prevailing wind. A certain proportion (29 %), however, was observed to fly against the wind, apparently in response to low levels of airborne attractants of unknown origin. With baited olfactometers placed upwind from the launching platform, the pattern of flight changed significantly ($\chi^2 = 44.99$, $P < 0.01$); a lesser proportion of the beetles (30 %) flew with the wind, while the proportion which flew against the wind increased to 54 percent.

The results obtained on August 24 differed markedly from those recorded on the succeeding two days, being very similar to those obtained in the absence of olfactometers. This inconsistency is believed to be due to the inadvertent use on this day of unattractive host material, i. e., 1-day-old pine bolts, in the olfactometers. Of particular interest was the pronounced response of beetles in flying towards the olfactometers on August 25 and 26, when freshly-cut host material was placed in the olfactometers at the beginning of each day. On August 25, 77 percent of the beetles flew towards the sides of the arena near the olfactometers or onto the machines themselves, as did 62 percent on August 26. Although some of the responding beetles flew across the wind to reach source of attraction, the majority flew in a direct path against the prevailing wind.

It is possible that emerging beetles perceive volatile attractants from nearby sources even before departing from brood trees. Indeed, when left undisturbed on the launching platform, the beetles we tested displayed a marked tendency to walk towards the edge of the platform closest to the olfactometers and to orient into the wind before taking flight. This observation, together with the results of flight tests, provide substantial evidence that *D. ponderosae* is capable of responding to olfactory stimuli immediately upon emergence without requiring prior flight exercise; an ability apparently not shared by either *D. pseudotsugae* (BENNETT and BORDEN 1971) or *Trypodendron lineatum* (Oliver) (BENNETT and BORDEN 1971; GRAHAM 1959).

A relatively greater proportion of females (32 and 68 %) than males (26 and 55 %) flew against the wind over the periods August 17–21 and August 25–26, respectively (Table 2). The differential response between the sexes, however, was not consistent from day to day. As a result, there was no significant difference ($\chi^2 = 3.20$, N. S. at 0.05) in the overall flight responses between males and females for the 8 days of observation. In related studies, GARA (1963) found that female *I. confusus* were superior in response to attractants over distances greater than 25 meters, while males responded in larger proportions over lesser distances.

The diurnal variation in initial flight response in relation to wind direction was considerable, particularly in the absence of baited olfactometers. Periodic changes in the direction of prevailing winds are believed to be responsible for much of this variation, since levels of natural attractants in the air no doubt differed with each wind change. The changes in wind direction experienced during each day were another factor worthy of mention. Even during the relatively short flight of beetles within the arena, several changes in wind direction occasionally occurred. The response of the beetles to these changes was notable; of the 30 or so beetles that changed their flight direction (Fig. 3), several did so in response to a change in wind direction. Such a reversal in flight direction is not as likely to be observed in smaller arenas, as for example the one described by SEYBERT and GARA (1970).

The heights which tested beetles attained upon arrival at the sides of the

arena did not appear to correlate with either temperature or wind speed (Fig. 4). In contrast, SEYBERT and GARA (1970) found that *I. pini* (Say), when released in a small arena, flew higher at low wind speeds and higher temperatures. It is possible that the ability to fly against high winds may vary with the sex of individual beetles; more males (30) than females (14) were recorded in flight at wind speeds in excess of 4 km/h (Fig. 3B).

Flight in an Open Situation:

In comparison to arena tests, in the open a smaller proportion (18 %) of beetles flew to attractants directly against the wind (Table 3). The proportion of beetles which flew across wind, however, was considerably greater in the open (25 %) than within the arena (15 %). These discrepancies are believed to result from the higher wind speeds that prevailed outside the arena. Although response to attractants was more predictable at low wind speeds (under 5 km/h), 11 beetles were observed to fly against winds exceeding 7.5 km/h.

Conclusion on Flight Behaviour: The flight studies described herein suggest that newly-emergent *D. ponderosae* adults tend

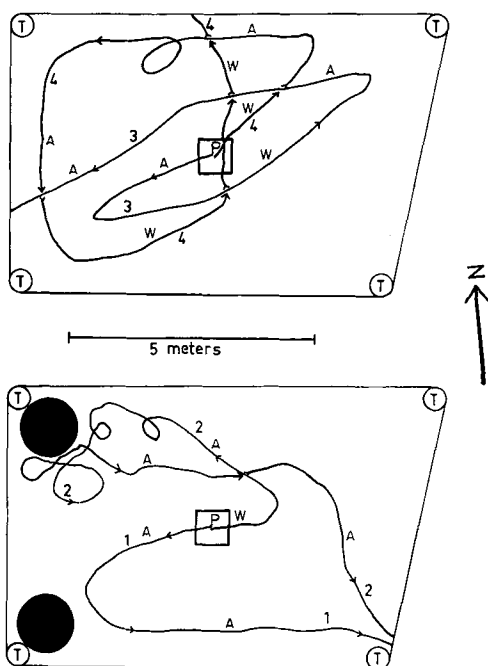


Fig. 3. Flight path of four *Dendroctonus ponderosae* beetles inside the arena in relation to directional changes in wind: T – corner tree, ● – olfactometer, P – launching platform, W – flight with the wind, A – flight against the wind

Table 3

Initial flight response of newly emergent *Dendroctonus ponderosae* beetles to wind when tested in an open situation

Date (1971)	Sex		With Wind	Flight Response		Total
				Against Wind	Across Wind	
August 24	♀		13	2	0	15
	♂		12	2	1	15
25	♀		11	2	3	16
	♂		8	2	3	13
26	♀		36	17	24	77
	♂		25	9	15	49
Total	♀		60	21	27	108
	♂		45	13	19	77
Total Grand	♀ + ♂		105	34	46	185

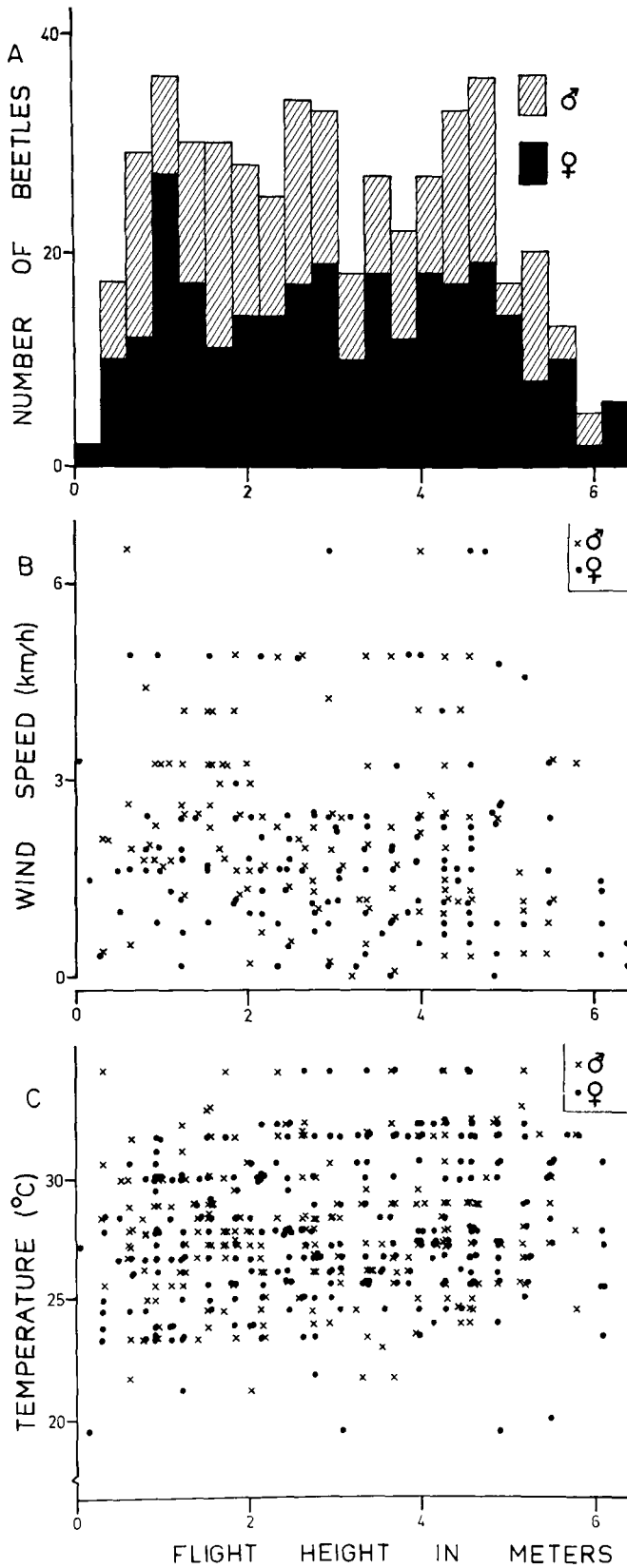


Fig. 4. Number of female and male *Dendroctonus ponderosae* beetles, wind speed and temperature at time of flight correlated against flight height on reaching the sides of the arena. (August 17-26.) Where the recorded wind speed fluctuated markedly during the flight of a beetle, the data have been excluded from Figure 4 B

to disperse with the wind in the absence of attractive host and insect produced volatiles. In the presence of attractive material, beetles respond readily against low prevailing winds without extensive flight exercise. Similar observations have been made for *I. confusus* (GARA 1963), *D. brevicornis* Led. (VITÉ and GARA 1962), and *D. frontalis* Zimm. (COSTER and GARA 1965). Differences in the ability to respond was found to occur individually and possibly between the sexes. That a significant decrease in the ability to respond to attractants as the flight season progresses has been shown for *I. confusus* in both the laboratory (BORDEN 1967) and the field (GARA 1963). Since our flight studies were conducted late in the flight season, it is not known whether a greater proportion of early emerging *D. ponderosae* broods are capable of olfactory response.

The fact that a significant proportion of both males and females in present experiments failed to respond to nearby attractants indicates the need for studies on population quality. Perhaps a certain proportion of the population is incapable of responding to secondary attractants, or require flight exercise before responding. Such individuals may function as pioneers which disperse into more distant areas to establish new infestation foci.

Acknowledgements

The authors wish to thank Mr. ORRIN GREEN, District Administrator of the Ellensburg District, State Department of Natural Resources, for the cooperation and help he and his staff provided. We also appreciate the advice and cooperation given by Dr. G. B. PITMAN of Boyce Thompson Institute, Grass Valley, California.

Zusammenfassung

Über das Schlüpfen und Anfangsflugverhalten des Borkenkäfers Dendroctonus ponderosae, in Ost-Washington

Das Schlüpfen von *D. ponderosae* aus der Kiefer sowie das Anfangsflugverhalten der Käfer wurden unter Freilandbedingungen untersucht. Die stündliche Schlüpfquote stieg zwischen 20 und 30° C mit der Temperatur an, sank aber bei höherer Temperatur deutlich ab. Unter 16° C fand praktisch kein Schlüpfen mehr statt. Im optimalen Temperaturbereich zwischen 25 und 30° C war die Spitze des stündlichen Schlüpfens nicht mit dem Maximum der Tagestemperatur synchronisiert, woraus hervorgeht, daß ein bestimmter Schlüpfrrhythmus besteht. Das mittlere Geschlechterverhältnis schlüpfender Käfer betrug im Mittel 1:1,4 zugunsten der ♀♀. Bei Versuchen flogen frisch geschlüpfte Käfer im allgemeinen mit dem Wind, wenn keine Duftstoffe vorhanden waren. Dagegen flogen sie bei Anwesenheit von synthetischem trans-Verbenol und frischem Brutmaterial zumeist gegen den Wind direkt auf die Anlockstoffe zu. Für die Ausübung des Fluges bildete jedoch eine Duftanlockung keine Voraussetzung. Die Flughöhe über dem Erdboden schien unabhängig von der Temperatur und Windgeschwindigkeit zu sein. Jedoch nahm der Anteil der auf Anlockstoffe ansprechenden Käfer mit zunehmender Windgeschwindigkeit ab. Einige Käfer flogen noch bei einer Windgeschwindigkeit von 7,5 km/h gegen den Wind.

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